# Weekly Progress Report

**Internship:** Free Summer Internship – Upskill Campus
**Name:** MUKUL VERMA **Date:** 28/06/2026

## Project 1: Crop and Weeds Detection using YOLOv8

### 1. FORMAT

Report prepared in an organized manner with proper headings, subheadings, and bullet points as per the internship guidelines.

### 2. CONTENT

**Tasks Completed:**

- Created a custom dataset of crops and weeds and split it into Train, Validation, and Test sets.
- Configured the `data.yaml` file and trained the YOLOv8 object detection model.
- Generated the trained weights file (`best.pt`).
- Evaluated the model using the `model.val()` function.
- Achieved mAP50: 0.87, Precision: 0.89, and Recall: 0.85 on 260 validation images.
- Performed object detection on test images using `model.predict()`.
- Implemented real-time webcam detection with approximately 7 ms inference time.

**Milestones Achieved:**

- Successfully trained a custom YOLOv8 model with approximately 87% detection accuracy.
- Completed the entire object detection pipeline from dataset preparation to real-time deployment.

### 3. CHALLENGES AND HURDLES

- **Format Incompatibility:** Faced format mismatch issues with raw annotation labels, which required writing a custom Python script to convert them seamlessly into standard YOLO format.
- **Accuracy Bottlenecks:** Initial training runs yielded lower detection confidence, which was resolved by fine-tuning the model through additional training epochs and applying data augmentation techniques.
- **Hardware Limitations:** Training locally was highly time-consuming due to standard computational limits; bypassed this bottleneck by migrating the pipeline to a Google Colab GPU environment to significantly accelerate training.

### 4. LESSONS LEARNED

- Understood the distinct operational differences between `model.val()` (evaluation) and `model.predict()` (inference).
- Learned core object detection evaluation metrics including Precision, Recall, and mAP50.

• Gained practical, hands-on experience in dataset preparation, structural model training, and real-time object detection.

• Understood the critical real-world importance of computer vision and AI applications in precision agriculture.

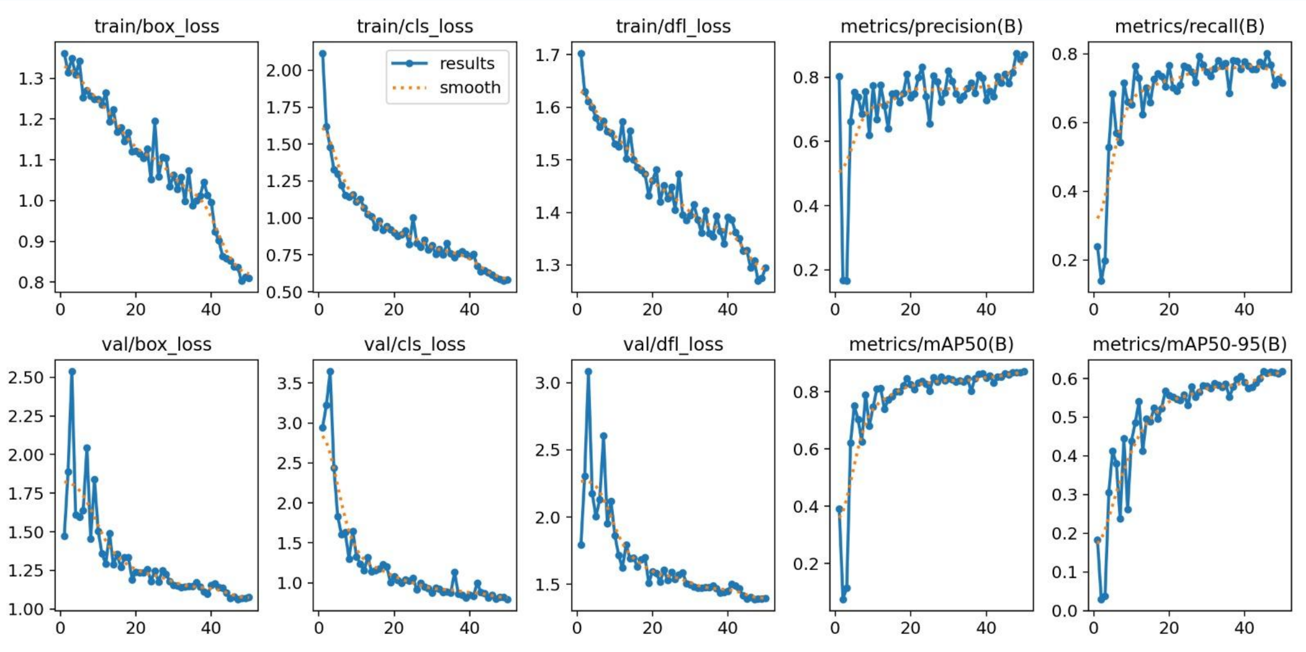


**Figure 1: YOLOv8 Training Results (Loss and Metrics Curves)**

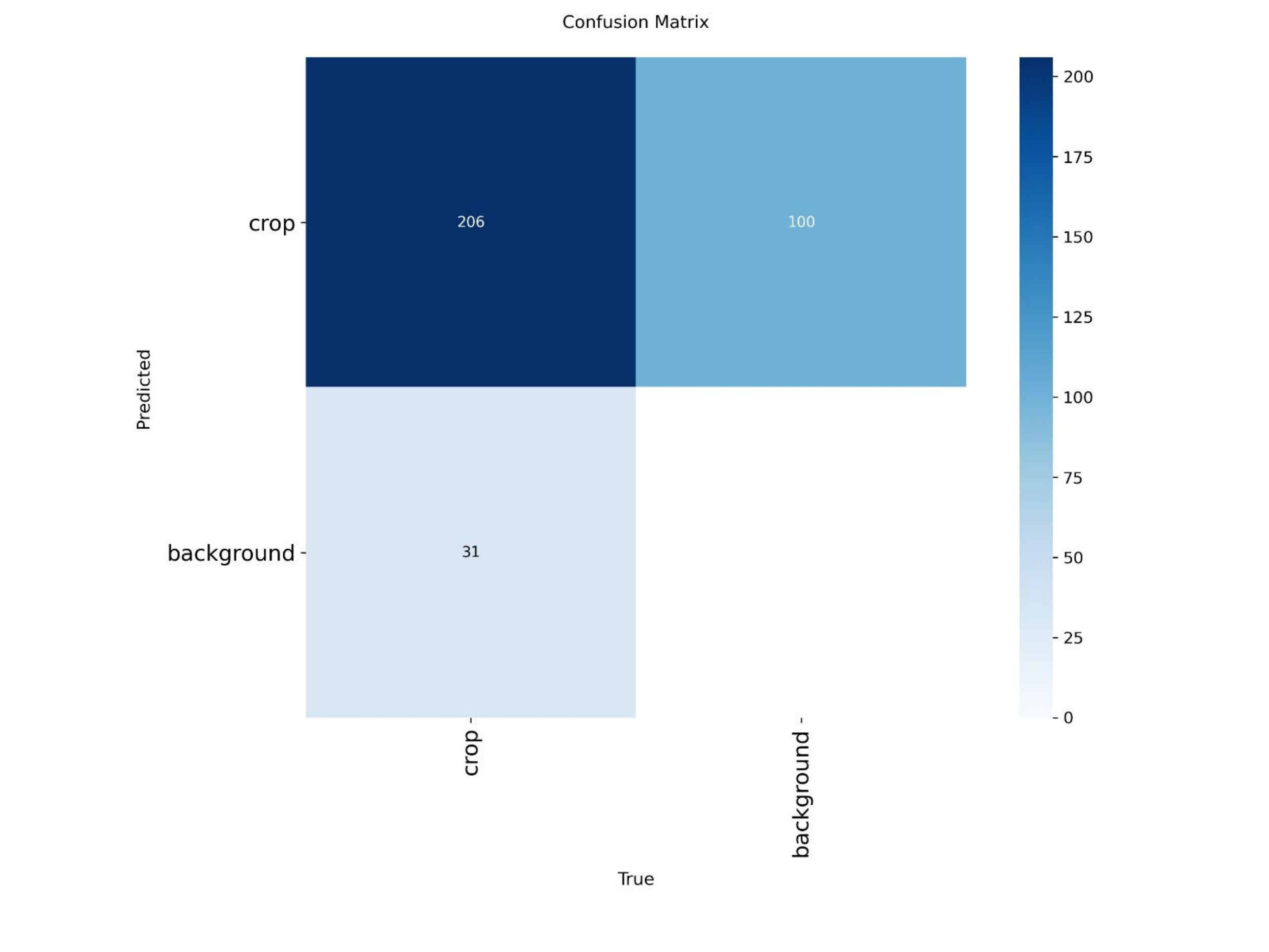


**Figure 2: Model Confusion Matrix**

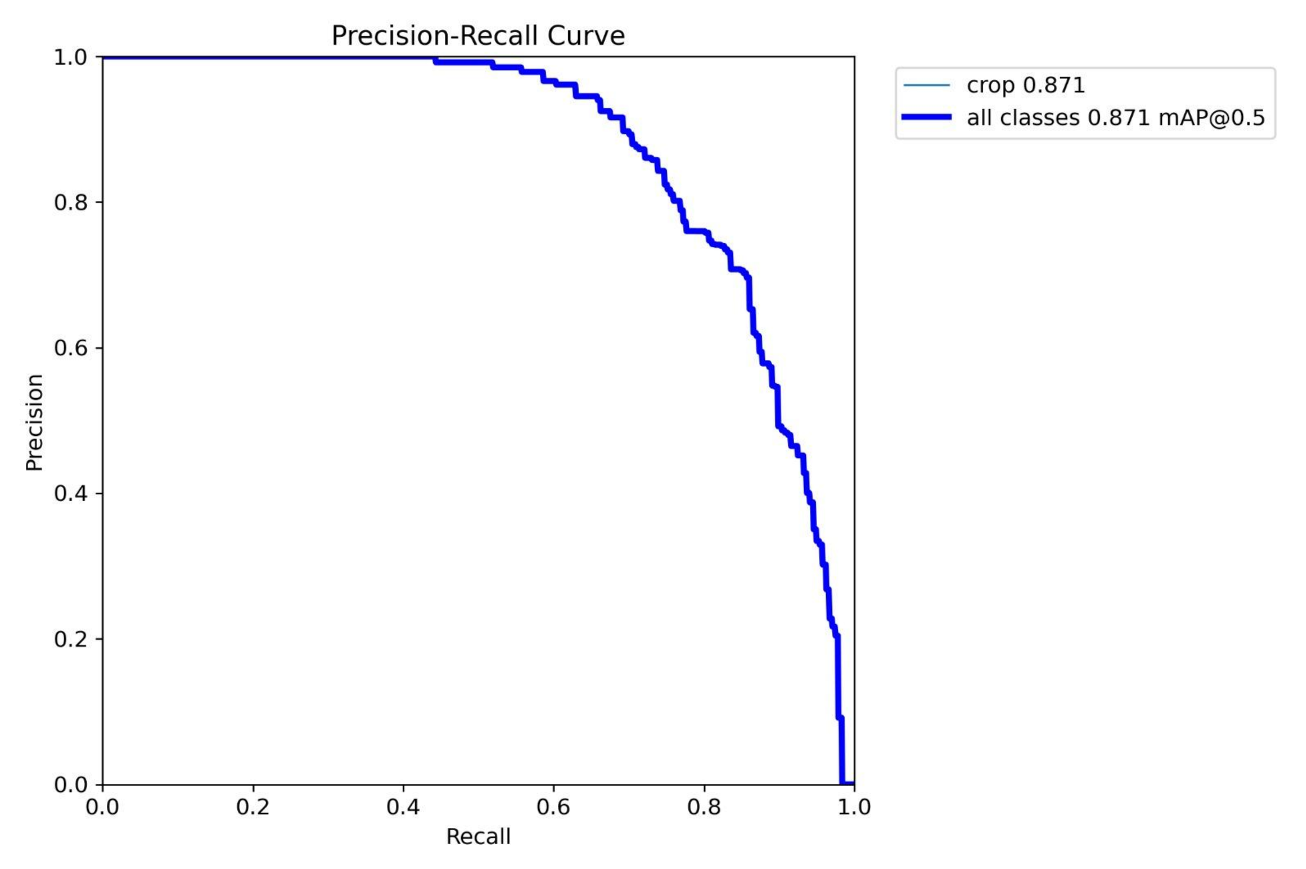


**Figure 3: Model Evaluation Metrics (Precision-Recall Curve)**

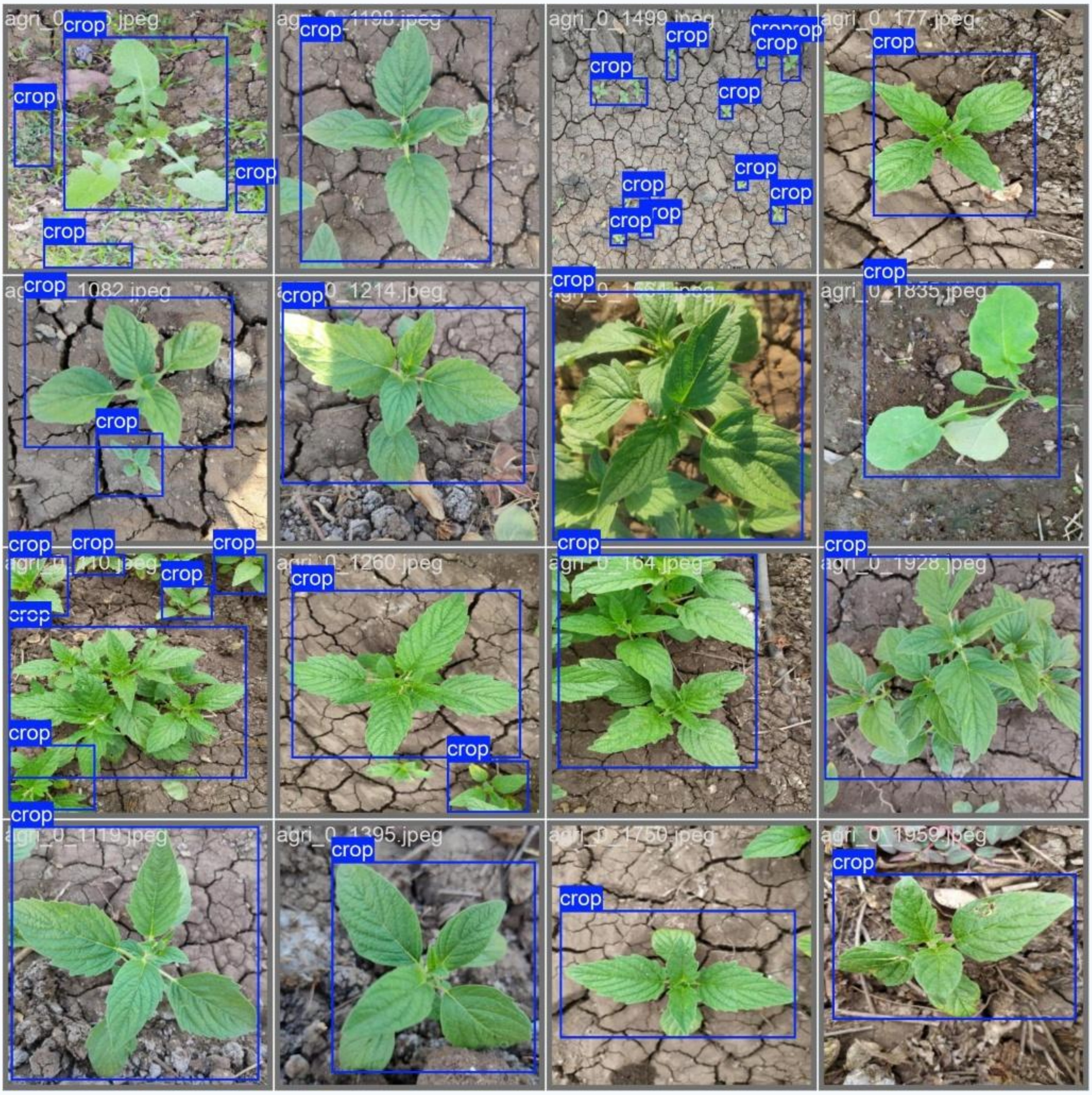


**Figure 4: Sample Crop and Weed Validation Detections**

## Project 2: Traffic Forecasting using Facebook Prophet

### 1. FORMAT

Report prepared using professional formatting with clear headings and bullet points.

### 2. CONTENT

**Tasks Completed:**

- Loaded and preprocessed hourly traffic data from 2015–2017.
- Trained the Facebook Prophet forecasting model.
- Enabled daily, weekly, and yearly seasonalities.
- Generated traffic forecasts until 2027.
- Analyzed Trend, Daily, Weekly, and Yearly seasonal components.
- Predicted nearly 20% traffic growth by 2027.

**Milestones Achieved:**

- Successfully implemented a complete time-series traffic forecasting pipeline.
- Identified distinct peak traffic hours (10 AM–12 PM and 6 PM–8 PM).
- Determined that Sundays experience the lowest overall traffic volume.

### 3. CHALLENGES AND HURDLES

- Resolved strict Prophet installation and library dependency issues in the environment setup.
- Filled missing hourly data records using forward-fill interpolation to prevent sequence gaps.
- Developed the analytical skills required to accurately interpret forecast trends and seasonal component graphs.

### 4. LESSONS LEARNED

- Learned practical time-series forecasting frameworks using Facebook Prophet.
- Understood trend analysis methodologies and seasonal decomposition.
- Learned the statistical importance of utilizing confidence intervals in predictive forecasting.
- Gained insight into how predictive traffic data actively supports smart city infrastructure planning.

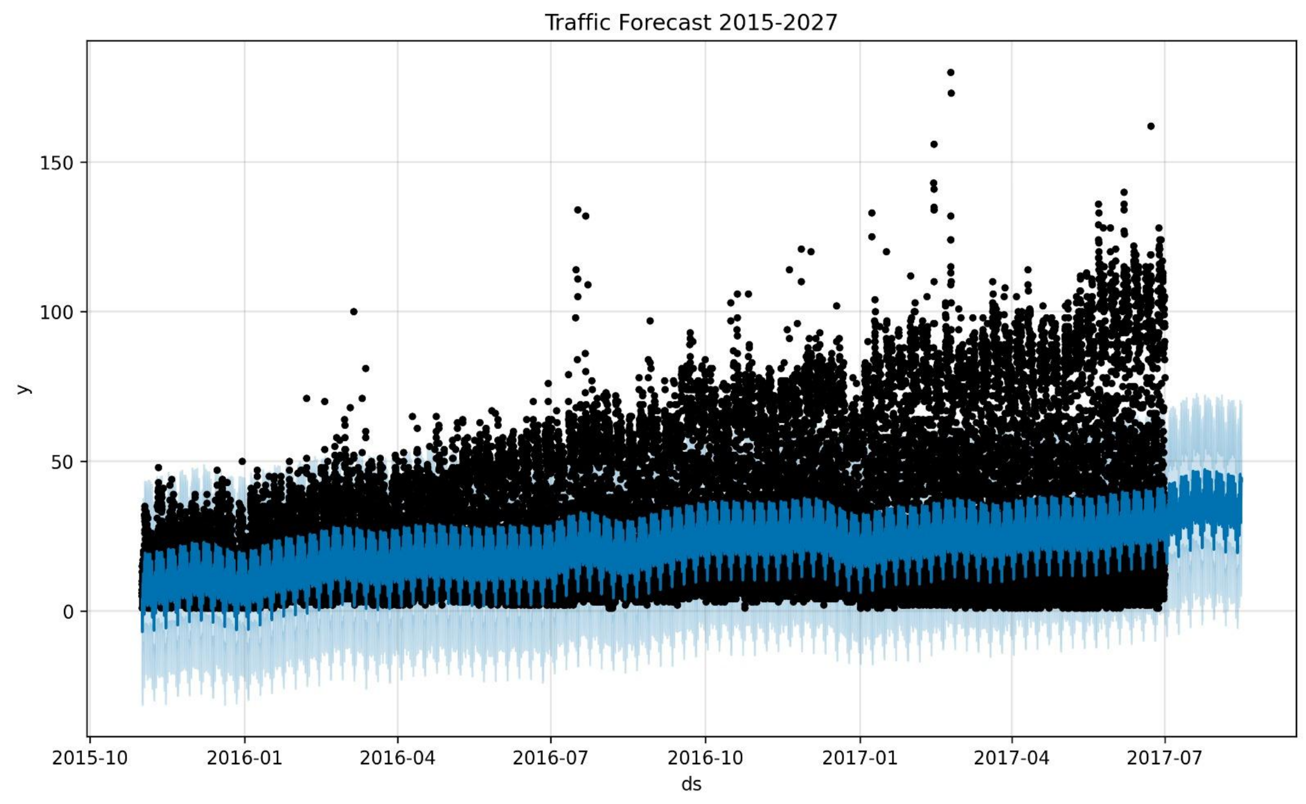


**Figure 5: Traffic Forecast Graph (2015–2027)**

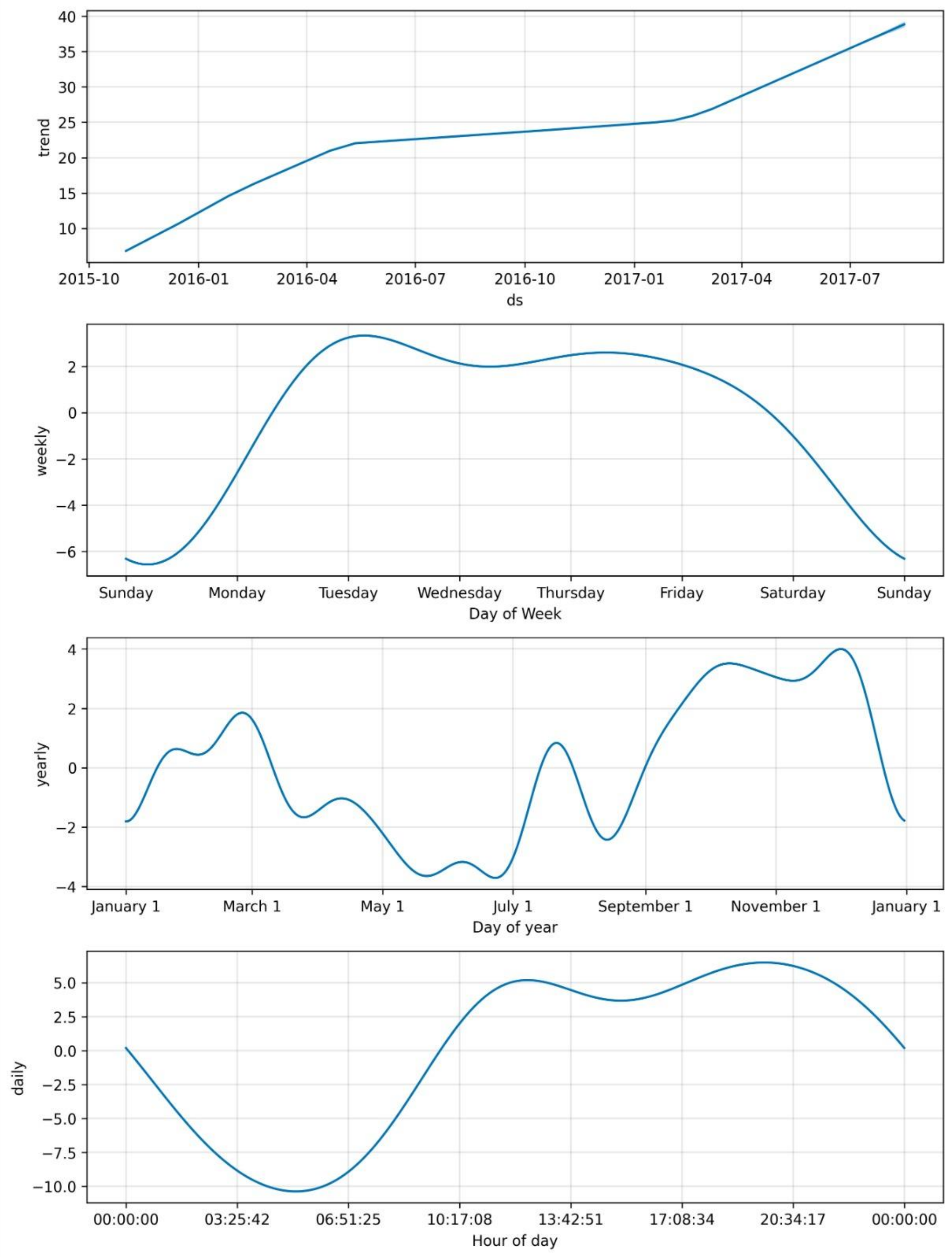


**Figure 6: Prophet Seasonal Decomposition Components**